Exam 2

ECON 306 — Intermediate Microeconomic Theory

Exam 2, Summer Session I 2026

Instructions

- Write the answer to questions on a separate sheet of paper.
- DO NOT WRITE YOUR ANSWERS ON A TABLET OR IPAD. USE OF THOSE DEVICES IS STRICTLY PROHIBITED DURING THE EXAM.
- Each question is 1 point unless otherwise marked.
- Answers without appropriate work may not receive full credit so please show your work!
- Partial credit may be given but it is up to my discretion.
- Unless otherwise specified, assume preferences are well behaved.
- Round answers if needed to the nearest 1/100th (0.01).
- A calculator may be used.
- Use of notes or any resource (including AI) is STRICTLY prohibited.

Scenario A

Consider a competitive market for steel with the following inverse demand and inverse supply curves:

$$P = 60 - 0.2Q \qquad \text{(Inverse Demand)}$$

$$P = 10 + 0.3Q \qquad \text{(Inverse Supply)}$$

where Q is the quantity of steel produced in tons and P is the price per ton. The production of steel also creates a negative externality given by the following Marginal External Cost (MEC) function:

$$MEC = 5 + 0.1Q.$$

1. (2 points) Find the producer and consumer surplus in the unregulated steel market.
2. (2 points) What is the deadweight loss resulting from the externality?
3. Suppose the government wishes to eliminate the deadweight loss through imposing a tax on steel. What kind of tax should the government use?
4. (2 points) Find the tax amount (per ton) that eliminates the deadweight loss and the government revenue from the tax.
5. (2 points) What will be the consumer price and consumer surplus with the tax?

Scenario B

The land of Ashina is known for its Rejuvenating Waters that are produced locally by a single manufacturer: Genichiro. The demand for the Waters is given by the following inverse demand:

$$p = 1000 - 2Q.$$

To produce the water, there is a fixed cost of 50,000 and a constant marginal cost of 40.

6. (3 points) Acting as a monopolist, what price and quantity of water will Genichiro set? Find their profits.

7. Calculate the Lerner Index (price-cost markup) at the monopoly price and quantity.

8. What is the deadweight loss created by the monopoly?

Scenario C

Consider the game played between two firms. Firm 1 chooses the row and Firm 2 chooses the column; each cell lists (Firm 1's payoff, Firm 2's payoff).

		Firm 2		
		Low	Med	High
	Low	25, 18	12, 30	8, 22
Firm 1	Med	30, 12	22, 22	16, 14
	High	27, 27	14, 8	11, 5

9. What is player 1's dominant strategy (if any)?

10. What is player 2's dominant strategy (if any)?

11. (2 points) What is (are) the pure Nash equilibrium(s) if any? Write out all the pure NE below or write "none" if there are none.

Scenario D

A taxpayer decides whether to *Evade* or *Report* their income honestly, while the tax authority decides whether to *Audit* or *Not Audit*. Payoffs are (Taxpayer, Auditor):

		Auditor	
		Audit	No Audit
	Evade	0, 4	10, 0
Taxpayer	Report	5, 1	5, 3

12. (3 points) Show that this game has no pure-strategy Nash equilibrium, then find the mixed-strategy Nash equilibrium: the probability p that the taxpayer Evades and the probability q that the auditor Audits.

Scenario E

Consider a Cournot Duopoly where the inverse market demand is

$$p = 500 - 5Q,$$

where $Q = q_1 + q_2$. Each firm has the cost function $C(q) = 50q$.

13. (2 points) Find the Cournot equilibrium output of each firm.

14. (2 points) Suppose the firms form a cartel to maximize their combined profits. Assuming they each produce an equal quantity and are able to stick to the promised output, what output does each firm produce?

15. What best describes the difficulty in sustaining a cartel between these two firms?

    A. Each firm can make a greater profit than their share of the cartel profit by producing a greater amount of output.

    B. Each firm can make a greater profit than their share of the cartel profit by producing a smaller amount of output.

    C. Each firm makes more profit from the Cournot equilibrium outputs than the Cartel outputs.

    D. Firms will each want to produce the perfect competition output level so as to not risk losing money.

16. Continuing the scenario from above, suppose firm 1 believes that firm 2 will produce the agreed upon cartel level of output. What output would firm 1 choose to produce?

Multiple-Choice Questions

17. The monopoly price-cost markup will be greater for markets with a ______________ demand.
    A. Less elastic
    B. More elastic
    C. Perfectly elastic
    D. There is no relationship between elasticity and the price-cost markup
18. If the demand function for a monopoly's product is $Q = 40 - 0.25p$, then the firm's marginal revenue function is
    A. $160 - 4Q$
    B. $160 - 8Q$
    C. $40 - 8Q$
    D. $40 - p$
19. In markets where positive externalities are present, a ______________ that is set equal to the ______________ will increase the market output to its efficient level.
    A. Tax; social marginal cost
    B. Tax; marginal external cost
    C. Subsidy; private marginal benefit
    D. Subsidy; marginal external benefit
20. A competitive market has demand $P = 80 - 2Q$ and supply $P = 20 + Q$. At the market equilibrium, the consumer surplus is:
    A. 400
    B. 200
    C. 600
    D. 800
21. According to the Coase Theorem, when property rights are clearly defined and the parties can bargain at negligible transaction cost, private bargaining will:
    A. reach an efficient outcome only if the injured party is assigned the rights
    B. reach an efficient outcome regardless of which party is initially assigned the property rights
    C. fail to internalize the externality without a corrective tax
    D. always require government intervention to be efficient

22. In the long run, a firm in a perfectly competitive market:
    - A. earns positive economic profit protected by barriers to entry
    - B. produces where price equals the minimum of average total cost and earns zero economic profit
    - C. sets price above marginal cost to maximize profit
    - D. shuts down whenever price falls below average total cost

23. Two firms produce a homogeneous good with identical constant marginal cost c and compete by simultaneously setting prices (Bertrand competition). In the unique Nash equilibrium:
    - A. both firms set the monopoly price and split the profit
    - B. both firms set price equal to marginal cost and earn zero profit
    - C. each firm earns its Cournot-equilibrium profit
    - D. price exceeds marginal cost because only two firms compete

24. Which of the following best describes when a firm should "shut down" in the short run?
    - A. When a firm's revenues are less than the fixed costs
    - B. When a firm's revenues are less than the variable costs
    - C. When a firm's profits are negative
    - D. When a firm's profits are less than fixed costs

25. Which of the following is NOT a characteristic of a perfectly competitive market?
    - A. There are many buyers and sellers
    - B. There are no transaction costs
    - C. Firms sell a homogeneous good
    - D. Each firm faces a downward sloping demand curve

Exam 2 — Formula Sheet

Elasticities

$$E = \frac{\partial Q}{\partial p} \cdot \frac{p}{Q}, \qquad E_D = \frac{\partial Q_D}{\partial p} \cdot \frac{p}{Q_D}, \qquad E_S = \frac{\partial Q_S}{\partial p} \cdot \frac{p}{Q_S}, \qquad E_Y = \frac{\partial Q_D}{\partial Y} \cdot \frac{Y}{Q_D}$$

Costs

$$AC = \frac{C(q)}{q}, \qquad MC = C'(q)$$

Profit

$$\pi = TR - TC = pq - c(q) = q\,(p - AC), \qquad \text{Profit max where } MR = MC$$

Producer Surplus

$$PS = TR - VC = q\,(p - AVC) \quad \text{(single firm)}$$

$$PS = \text{area below price and above Supply} \quad \text{(market)}$$

Consumer Surplus

$$CS = \text{area below demand and above price} = \text{Total Value} - \text{Total Expenditure}$$

Lerner Index (price-cost markup)

$$L = \frac{p - MC}{p}$$

Monopoly Optimal Price-Cost Markup

$$L = \frac{p - MC}{p} = \frac{1}{|E_D|}$$

Area of a Trapezoid

$$A = \frac{1}{2}\,(b_1 + b_2)\,h$$

Production

$$Q = f(K, L) \text{ (Total Product)}, \qquad MP_L = \frac{\partial Q}{\partial L}, \qquad MP_K = \frac{\partial Q}{\partial K}, \qquad AP_L = \frac{Q}{L}$$

$$MRTS = -\frac{MP_L}{MP_K} \text{ (slope of isoquant)}, \qquad \text{slope of isocost} = -\frac{w}{r}$$

Examples of Production Functions

$$\text{Perfect Substitutes:} \quad f(K, L) = aK + bL, \qquad a > 0,\ b > 0$$

$$\text{Fixed Proportions:} \quad f(K, L) = c\,\min\{aK,\, bL\}, \qquad a, b, c > 0$$

$$\text{Cobb–Douglas:} \quad f(K, L) = A\,K^a L^b, \qquad A, a, b > 0$$

$$\text{Cobb–Douglas } MRTS = -\frac{MP_L}{MP_K} = -\frac{b\,K}{a\,L}$$

Cost of Production

$$\text{Total cost of inputs} = \text{Labor cost} + \text{Capital cost} = wL + rK$$

$$C(Q) \text{ (Total Cost)}, \qquad AC(Q) = \frac{C(Q)}{Q}, \qquad MC(Q) = C'(Q)$$

Short-Run Cost Curves

$$C(Q) = VC(Q) + FC, \qquad AVC(Q) = \frac{VC(Q)}{Q}, \qquad AFC(Q) = \frac{FC}{Q}$$

Production and Cost Relationship

$$MC = \frac{w}{MP_L}, \qquad AVC = \frac{w}{AP_L}$$